FAST-NU
Lahore Campus
Department of Electrical Engineering
Dated: Tuesday, September 9, 2025

EE3005 – Electromagnetic Theory
Fall 2025
Section: BEE-5C
Quiz # 1 on CLO # 1

Name: _____

Roll No.: _____

Total Marks: 20

Time Allowed: 20 minutes

[20 marks]

Q1 The surfaces $r = 0$ *and* $1, \theta = 0°$ *and* $35°, \varphi = 0°$ *and* $305°$ identify a closed surface.

(a) **Identify** the coordinate system and name all the surfaces/planes formed by the respective coordinate variables with their ranges. Also, draw the legend of RCS as is superimposed on SCS shown in Figure 1. Fill below. **[5]**

(b) **Find** the total volume enclosed. **[5]**

(c) **Find** surface area along a_r only (also define a_θ uniquely). **[5]**

(d) **Find** the distance between $P(r = 1, \theta = 35°, \varphi = 0°)$ *and* $Q(r = 1, \theta = 35°, \varphi = 305°)$ on the ***circular arc***. **Identify** these points on the diagram as well.

CLO 01

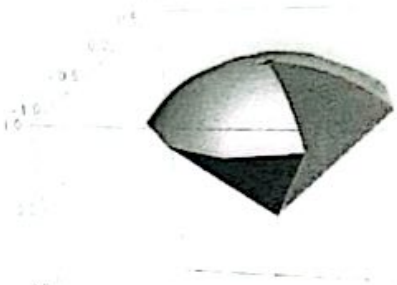


Figure 1. A 3D shape with filled surfaces. Show the legend of RCS as well.

Name of the Coordinate System:			
Name of the Surfaces / Planes formed:	$r = const.$ is		
	$\theta = const.$ is		
	$\varphi = const.$ is		
Ranges of the Coordinate Variables:	r	θ	φ

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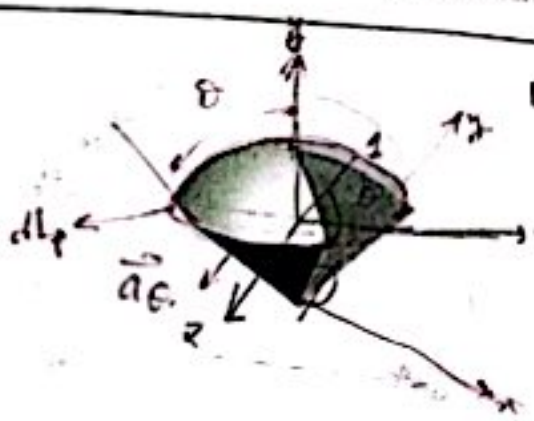


Figure 1. A 3D shape with filled surfaces. Show the legend of RCS as well.

Name of the Coordinate System:	Spherical Coordinate		
Name of the Surfaces / Planes formed:	r = const. is a Sphere		
	θ = const. is a cone		
	φ = const. is a half plane		
Ranges of the Coordinate Variables:	r	θ	φ
	0 to 1	0 to 35°	0 to 305

b) $V = \iiint_v r^2 \sin\theta \, dr \, d\theta \, d\varphi$ (limits: r: 0 to 1; θ: 0 to $\frac{7\pi}{36}$; φ: 0 to $\frac{61\pi}{36}$)

$= \int_0^{\frac{61\pi}{36}} \int_0^{\frac{7\pi}{36}} \int_0^1 r^2 \sin\theta \, dr \, d\theta \, d\varphi$

$= \int_0^{\frac{61\pi}{36}} \int_0^{\frac{7\pi}{36}} \frac{r^3}{3}\Big|_0^1 \sin\theta \, d\theta \, d\varphi$

$= \frac{1-0}{3} \left[\frac{61}{\pi}\right] \left[-\cos\left(\frac{7\pi}{36}\right) + \cos 0\right]$

$= \frac{61}{108} \left(1 - \cos\frac{7\pi}{36}\right)$

$V = 0.3208 \, m^3$

c) S along arc

$dS = r^2 \sin\theta \, d\theta \, d\varphi$ ~ $r = 1$

$dS = \sin\theta \, d\theta \, d\varphi$

$ds = \int_0^{\frac{7\pi}{36}} \int_0^{\frac{61\pi}{36}} \sin\theta \, d\theta \, d\varphi$

$= 2 \int_0^{\frac{61\pi}{36}} d\varphi \int_0^{\frac{7\pi}{36}} \sin\theta \, d\theta$

$= \frac{61\pi}{36} \left[-\cos\theta\right]_0^{\frac{7\pi}{36}}$

$= \frac{61\pi}{36} \left[1 - \cos\left(\frac{7\pi}{36}\right)\right]$

$A = 0.9626 \, m^2$

d) As $r = dr$

$a_\theta = r \, d\theta$

$\varphi = r \sin\theta \, d\varphi$

during circular moment $d\theta = 0$

So arc length $= r \sin\theta \, d\theta$

$= 1\left(\sin\left(\frac{7\pi}{36}\right)\right)\frac{61\pi}{36}$

$= 3.056$

$L_4 = \int dL_4 = \int_{305}^{61\pi/36} r \sin\theta \, d\varphi$

CamScanner

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